A PROJECT REPORT ON

STOCKSENSE: A SELF-EXPLAINING, EVENT-SOURCED INVENTORY RECONCILIATION ENGINE FOR MULTI-CHANNEL SELLERS

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING (Computer Engineering)

BY

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STOCKSENSE: A SELF-EXPLAINING, EVENT-SOURCED INVENTORY RECONCILIATION ENGINE FOR MULTI-CHANNEL SELLERS

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STOCKSENSE: A SELF-EXPLAINING, EVENT-SOURCED INVENTORY RECONCILIATION ENGINE FOR MULTI-CHANNEL SELLERS

(StockSense: A Self-Explaining, Event-Sourced Inventory Reconciliation Engine for Multi-Channel Sellers)

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Abstract

Modern retail and e-commerce merchants frequently list identical Stock Keeping Units (SKUs) simultaneously across multiple independent sales channels, including online marketplaces such as Amazon and Flipkart, direct-to-consumer (D2C) web storefronts, and physical point-of-sale (POS) shop counters. Because each sales channel operates as an isolated system with its own local view of inventory, keeping stock counts synchronized in real time is a critical operational challenge. Existing budget-friendly inventory tools rely on periodic polling intervals of 15 to 30 minutes and destructive state-overwrite updates. This architectural limitation creates five persistent failure modes: synchronization blind spots between polling cycles, concurrent race conditions when two channels sell the final unit simultaneously, double-deduction of stock caused by duplicate webhook retries during network latency spikes, naive first-come-first-served or negative-stock failure states during stock-outs, and complete absence of a permanent audit trail explaining why a stock discrepancy occurred. Conversely, enterprise Order Management Systems (OMS) such as Cin7 require expensive monthly subscriptions ($350 to $1,000+ per month) and rely on rigid, manually configured static rules, while standalone profit-analytics platforms (such as TrueProfit) operate in isolation from real-time inventory allocation decisions.

To bridge this gap for small and mid-scale multi-channel sellers, this project presents the **Multi-Channel Inventory Reconciliation Engine**, a fault-tolerant, event-sourced distributed system engineered to maintain a single, verifiable source of truth across heterogeneous sales channels. Instead of overwriting a single mutable stock counter, the engine applies the *Event Sourcing* architectural pattern: every sale, return, restock, and cancellation is permanently recorded as an immutable domain event in a PostgreSQL ledger, and current stock is derived deterministically as the cumulative fold of all historical events. Incoming channel webhooks are load-balanced through Nginx across three horizontally scaled Node.js and Express intake nodes, deduplicated in sub-millisecond time via atomic idempotency keys in Redis, and persisted into a durable RabbitMQ message queue to guarantee zero event loss even under server crashes.

When a Python reconciliation worker detects that concurrent sales across channels would drive stock below zero, the system executes a smart, priority-based conflict resolution algorithm. Rather than blindly rejecting an order or allowing negative stock, the engine evaluates competing orders using business-critical metrics—specifically net profit margin, customer importance, and channel cancellation penalties—to recommend or automatically unwind the least costly order while immediately notifying the affected customer and logging a permanent audit record. Furthermore, the project introduces a genuine architectural innovation absent from existing basic, enterprise, and analytics tools: an **adaptive, human-in-the-loop recommendation loop**. By continuously analyzing its own immutable history of stock-out conflicts and margin losses, the engine proactively surfaces evidence-based safety buffer recommendations (for example, holding back specific units on high-conflict channels) on a live HTML/JavaScript seller dashboard. Crucially, a human operator always approves or rejects each suggestion before any allocation policy changes, combining automated operational intelligence with strict human financial accountability.

Acknowledgments

It gives us great pleasure in presenting the project report on ***"StockSense: A Self-Explaining, Event-Sourced Inventory Reconciliation Engine for Multi-Channel Sellers"*** *prepared as part of the Bachelor of Engineering in Computer Engineering curriculum for the Academic Year 2026-2027.*

We would like to take this opportunity to express our deep sense of gratitude to our internal guide, ***Mr. Basappa Karlatthe****, Department of Computer Engineering, for providing continuous encouragement, technical insights, and constructive guidance throughout the formulation, architectural design, and documentation of this project. Their insightful feedback on distributed systems, event sourcing, and conflict resolution models has been invaluable.*

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Chapter 1

Synopsis

1.1 Project Title

StockSense: A Self-Explaining, Event-Sourced Inventory Reconciliation Engine for Multi-Channel Sellers

1.2 Project Option

Internal / Entrepreneurial Engineering Project — Designed to solve a high-impact real-world multi-channel retail and e-commerce infrastructure challenge for small and mid-size merchants operating across marketplaces (Amazon, Flipkart), direct websites, and brick-and-mortar POS counters.

1.3 Internal Guide

Mr. Basappa Karlatthe, Department of Computer Engineering, Dr. D.Y. Patil Institute of Technology, Pimpri, Pune.

1.4 Sponsorship and External Guide

Internal Academic Research and Development Project carried out at the Department of Computer Engineering, Dr. D.Y. Patil Institute of Technology, Pimpri, Pune (Self-Sponsored / Departmental Laboratory Infrastructure).

1.5 Technical Keywords (As per ACM Keywords)

Following the ACM Computing Classification System (CCS), the technical taxonomy of this project is structured as follows:

1. **C. Computer Systems Organization**
    (a) **C.2 Computer-Communication Networks**
        i. **C.2.4 Distributed Systems**
            A. Client/server and webhook-driven architectures
            B. Distributed applications and message-oriented middleware (AMQP / RabbitMQ)
            C. Distributed databases and immutable event-sourced ledgers
            D. Reliability, idempotency, and fault tolerance in distributed event streams
2. **H. Information Systems**
    (a) **H.2 Database Management**
        i. **H.2.4 Systems**
            A. Transaction processing and optimistic concurrency control
            B. Event Sourcing and Command Query Responsibility Segregation (CQRS)
3. **J. Computer Applications**
    (a) **J.1 Administrative Data Processing**
        i. Multi-channel e-commerce inventory synchronization
        ii. Automated conflict resolution and human-in-the-loop decision support

1.6 Problem Statement

To design and implement a centralized, fault-tolerant **Multi-Channel Inventory Reconciliation Engine** based on immutable Event Sourcing, Redis-backed idempotency deduplication, RabbitMQ durable message queuing, and Python-based priority conflict resolution that eliminates multi-channel overselling across independent retail platforms (Amazon, Flipkart, direct web store, and physical POS counter), preserves a complete audit trail of every stock mutation, provides integrated order-level profit analytics, and proactively generates evidence-based, human-approved inventory buffer recommendations from historical conflict patterns.

1.7 Abstract

- Multi-channel sellers listing identical inventory across Amazon, Flipkart, direct web stores, and physical counters frequently suffer from overselling when concurrent sales occur within seconds of one another.
- Conventional budget tools poll channels every 15–30 minutes and overwrite stock numbers in place, resulting in sync blind spots, race conditions, duplicate webhook deductions, naive order rejections, and zero historical auditability.

- The proposed system replaces state-overwrite storage with an immutable **Event Sourcing** ledger in PostgreSQL where current stock is computed as the exact mathematical fold of all recorded sales, returns, restocks, and cancellations.
- Incoming webhook traffic is distributed by Nginx across three Node.js + Express intake nodes, checked for duplicates in sub-millisecond time via Redis, and persisted in a durable RabbitMQ queue for crash-resilient processing.
- When concurrent last-unit sales trigger a negative-stock conflict, a Python reconciliation worker evaluates order profit margins and customer value to unwind the lower-priority order automatically while logging the full rationale.
- As its core innovation, the engine continuously analyzes its own conflict history to surface specific, evidence-based channel buffer suggestions on a unified live profit and stock dashboard, keeping a human operator in final control of every financial decision.

1.8 Goals and Objectives

- **Real-Time Multi-Channel Synchronization:** Eliminate the 15–30 minute polling blind spot by ingesting real-time sale, return, restock, and cancellation events via an Nginx-balanced cluster of three Node.js + Express webhook intake nodes.
- **Zero Duplicate Deductions (Idempotency):** Guarantee at-most-once stock deduction for retried channel notifications by enforcing atomic idempotency key checks in a fast Redis cache prior to queue insertion.
- **Crash-Resilient Event Durability:** Ensure zero event loss during traffic spikes or process failures by persisting validated events into a durable RabbitMQ message queue and an append-only PostgreSQL event store.
- **Margin-Aware Priority Conflict Resolution:** Detect concurrent stock-out race conditions deterministically and resolve conflicts by comparing net profit margin, customer lifetime importance, and channel SLA penalties rather than relying on naive first-come-first-served rejection.
- **Unified Auditability and Profit Visibility:** Provide a single live HTML/JS seller dashboard displaying current stock, full immutable event history, order-level net profit, and top customer behavior from the same underlying event stream.
- **Self-Improving Human-in-the-Loop Recommendations:** Analyze historical conflict frequencies per channel and SKU to surface actionable buffer allocation suggestions that require explicit human approval before taking effect.

1.9 Relevant Mathematics Associated with the Project

Let the Multi-Channel Inventory Reconciliation System be formally modeled as an algebraic 7-tuple:

$$\mathcal{S} = \langle \mathcal{I}, \mathcal{O}, \mathcal{E}, \Sigma, \Phi, \mathcal{C}_{\text{success}}, \mathcal{C}_{\text{failure}} \rangle \tag{1.1}$$

- **Input ($\mathcal{I}$):** A stream of asynchronous webhook notifications arriving from M independent sales channels $\mathcal{CH} = \{\text{Amazon}, \text{Flipkart}, \text{Website}, \text{POS}\}$. Each incoming notification is a tuple:

$$e_k = (\text{id}_k, s_k, c_k, \tau_k, \Delta q_k, m_k, u_k, t_k) \tag{1.2}$$

  where $\text{id}_k \in \Sigma^*$ is the unique channel notification identifier, $s_k \in \mathcal{SKU}$ is the product SKU, $c_k \in \mathcal{CH}$ is the originating channel, $\tau_k \in \{\text{SALE}, \text{RETURN}, \text{RESTOCK}, \text{CANCEL}, \text{UNWIND}\}$ is the event type, $\Delta q_k \in \mathbb{Z}$ is the signed stock change ($\Delta q_k < 0$ for sales, $\Delta q_k > 0$ for restocks/returns), $m_k \in \mathbb{R}_{\geq 0}$ is the net profit margin of the order, $u_k \in [0, 1]$ is the normalized customer importance score, and $t_k \in \mathbb{R}_+$ is the event timestamp.

- **Output ($\mathcal{O}$):**

  1. Reconciled available stock projection $Q_t(s)$ for every $s \in \mathcal{SKU}$ broadcast to all channels $c \in \mathcal{CH}$.
  2. Conflict resolution decision $d_k \in \{\text{COMMIT}, \text{UNWIND}(o_j)\}$ whenever a race condition occurs.
  3. Live profitability metrics $\Pi(s, c)$ and human-in-the-loop safety stock buffer recommendation $B_{\text{rec}}(s, c) \in \mathbb{Z}_{\geq 0}$.

- **Mathematical Formulation (Idempotency & Event Sourcing State Fold):** Let $\mathcal{D}_{k-1} = \{\text{id}_1, \text{id}_2, \ldots, \text{id}_{k-1}\}$ denote the set of processed event keys stored in Redis. The idempotency filter function f_{dedup} is defined as:

$$\mathbb{I}_{\text{new}}(e_k) = \begin{cases} 1, & \text{if } \text{id}_k \notin \mathcal{D}_{k-1} \\ 0, & \text{if } \text{id}_k \in \mathcal{D}_{k-1} \text{ (duplicate webhook ignored)} \end{cases} \tag{1.3}$$

  Instead of overwriting a mutable variable, the canonical stock $Q_t(s)$ at discrete step t is computed as the fold over the immutable PostgreSQL event ledger $\mathcal{L}_t(s)$:

$$Q_t(s) = Q_0(s) + \sum_{e_k \in \mathcal{L}_t(s)} \mathbb{I}_{\text{new}}(e_k) \cdot \Delta q_k \tag{1.4}$$

- **Conflict Detection and Priority-Based Resolution Formulation:** A stock-out conflict occurs at event e_k if the tentative post-event stock satisfies:

$$\tilde{Q}_t(s) = Q_{t-1}(s) + \Delta q_k < 0 \tag{1.5}$$

  Let $\mathcal{W}(s, \Delta t)$ be the set of concurrent contending orders within the race window Δt for the last available unit(s) of SKU s. For each contending order $o_i \in \mathcal{W}(s, \Delta t)$, the Python conflict engine evaluates the composite retention utility score $U(o_i)$:

$$U(o_i) = w_1 \cdot \widehat{m}(o_i) + w_2 \cdot u(o_i) + w_3 \cdot \psi(c_i) \tag{1.6}$$

where $\widehat{m}(o_i) \in [0,1]$ is the normalized net profit margin, $u(o_i) \in [0,1]$ is the customer loyalty/repeat-buyer score, $\psi(c_i) \in [0,1]$ is the channel SLA penalty weight, and $w_1 + w_2 + w_3 = 1$. The order selected for automatic unwinding and customer notification is the one minimizing retention utility:

$$o^*_{\text{unwind}} = \arg \min_{o_i \in \mathcal{W}(s,\Delta t)} U(o_i) \tag{1.7}$$

Upon unwinding o^*_{unwind}, a compensating event e_{comp} with $\Delta q = +|q(o^*_{\text{unwind}})|$ is appended to $\mathcal{L}_t(s)$, restoring $Q_t(s) \geq 0$ while preserving the full audit history.

- **Evidence-Based Buffer Recommendation Formulation:** Over an observation horizon T (e.g., 30 days), let $N_{\text{conflict}}(s,c)$ be the number of lost-margin conflict cancellations attributed to channel c for SKU s. When $N_{\text{conflict}}(s,c) \geq \theta_{\min}$, the system generates a human-reviewable buffer suggestion:

$$B_{\text{rec}}(s,c) = \left\lceil \gamma \cdot \frac{N_{\text{conflict}}(s,c)}{T} \cdot \bar{q}_{\text{race}}(s,c) \right\rceil \tag{1.8}$$

Subject to the strict human-governance invariant: active channel buffer $B_{\text{active}}(s,c)$ is updated to $B_{\text{rec}}(s,c)$ if and only if HumanApproval(B_{rec}) = TRUE.

- **Objects, Morphisms, and Functional Relations:**
  - *Objects:* `RawWebhookPayload`, `VerifiedEvent`, `LedgerEntry`, `StockProjection`, `ConflictResolutionRecord`, `BufferRecommendation`.
  - *Morphisms:* ϕ_{intake} : `RawWebhookPayload` $\rightarrow$ `VerifiedEvent` (via Redis deduplication); ϕ_{append} : `VerifiedEvent` $\rightarrow$ `LedgerEntry` (via RabbitMQ and PostgreSQL); $\phi_{\text{fold}} : \mathcal{P}(\texttt{LedgerEntry}) \rightarrow \texttt{StockProjection}$.
  - *Overloading:* The event fold operator $\oplus$ is overloaded across event types (`SALE` decrements available stock and increments gross revenue; `RETURN` increments inspectable stock and adjusts realized margin; `UNWIND` restores stock and records conflict mitigation cost).

- **Success Conditions ($\mathcal{C}_{\text{success}}$):**
  1. $\forall s \in \mathcal{SKU}$, reconciled stock $Q_t(s) \geq 0$ after conflict resolution.
  2. Zero duplicate stock deductions for identical id_k across all 3 intake nodes.
  3. 100% of state transitions are reconstructible from the PostgreSQL event log.

- **Failure Conditions ($\mathcal{C}_{\text{failure}}$):**
  1. Simultaneous exhaustion of both primary PostgreSQL storage and durable RabbitMQ disk persistence.
  2. Malformed webhook payload lacking mandatory SKU or channel signature fields (routed safely to Dead-Letter Queue without corrupting ledger state).

1.10 Names of Conferences / Journals where papers can be published

- **IEEE/ACM Conferences:** IEEE International Conference on Distributed Computing Systems (ICDCS) / IEEE International Conference on Cloud Computing in Emerging Markets (CCEM).
- **Conferences/Workshops in IITs:** National Conference on Communications (NCC) / IIT Bombay Systems and Software Engineering Symposium.
- **Central Universities or SPPU Conferences:** Savitribai Phule Pune University (SPPU) Post-Graduate & B.E. Computer Engineering Research Conference (cPGCON).
- **International Journals:** IEEE Access / ACM Transactions on Software Engineering and Methodology (TOSEM) / International Journal of Cooperative Information Systems.

1.11 Review of Conference/Journal Papers supporting Project idea

Table 1.1 summarizes ten foundational peer-reviewed papers and technical works supporting the architectural components of the Multi-Channel Inventory Reconciliation Engine.

1.12 Plan of Project Execution

The project execution across Academic Year 2026-2027 is structured into eight sequential phases managed via a milestone tracker (detailed in Chapter 5 and Annexure D):

1. **Phase 1 (July – August 2026):** Problem formulation, gap analysis of existing basic/enterprise/analytics inventory tools, and mathematical modeling of event sourcing and conflict cost functions.
2. **Phase 2 (August – September 2026):** Software Requirement Specification (SRS), database schema design for immutable PostgreSQL event logs, and architectural blueprint.
3. **Phase 3 (September – October 2026):** Implementation of Nginx load balancer, 3x Node.js + Express webhook intake nodes, and Redis atomic idempotency deduplication layer.
4. **Phase 4 (October – November 2026):** Durable RabbitMQ event queue integration, Term-I synopsis & design verification, and preliminary prototype demonstration.
5. **Phase 5 (December 2026 – January 2027):** Implementation of Python event-processing worker, event-sourced stock recalculation fold, and margin-aware priority conflict resolution engine.

Sr.	Authors & Year	Title / Core Contribution	Relevance to Our System
1	H. T. Kung & J. T. Robinson (1981), *ACM TODS* [1]	*On Optimistic Methods for Concurrency Control*: Introduced non-blocking read/validation/write phases for concurrent transactions.	Forms the theoretical basis for detecting concurrent last-unit sales without locking channel webhooks.
2	M. Fowler (2005), *Enterprise Application Architecture* [2]	*Event Sourcing Pattern*: Formalized storing all application state changes as an immutable sequence of domain events.	Core ledger design: stock is derived by folding historical events rather than overwriting a counter.
3	G. Young (2010), *CQRS Documents* [3]	*Command Query Responsibility Segregation (CQRS)*: Separated write-side event ingestion from read-side materialized projections.	Enables the same event log to power both stock sync and the live seller profit dashboard.
4	J. Kabbedijk et al. (2012), *LNCS Springer* [4]	*Variability and Scalability in Systems Applying CQRS and Event Sourcing*: Demonstrated horizontal scalability of event-driven architectures.	Validates decoupling 3 Node.js intake nodes from Python processing workers via RabbitMQ.
5	B. Erb & F. Hauck (2016), *ACM DEBS* [5]	*Log-based Fault Tolerance for Reactive Event-Sourced Systems*: Showed deterministic crash recovery via event replay.	Guarantees zero lost inventory updates even if an intake or worker container crashes mid-flight.
6	M. Kleppmann (2017), *O'Reilly Media* [6]	*Designing Data-Intensive Applications*: Analyzed idempotency keys, exactly-once semantics, and distributed logs.	Guides our Redis `SETNX` idempotency layer and PostgreSQL append-only audit design.
7	K. Overeem et al. (2021), *arXiv / Empirical Software Eng.* [7]	*An Empirical Characterization of Event Sourced Systems*: Studied 19 production event-sourced systems and auditability benefits.	Confirms that event sourcing eliminates silent state corruption and enables retrospective analysis.
8	S. Karuppuchamy (2025), *ResearchGate / Supply Chain* [8]	*Multichannel Inventory Management and Its Role in Fast Fulfillment*: Highlighted overselling costs across multi-platform retail.	Establishes the business urgency of real-time cross-channel reconciliation for mid-size sellers.
9	WJARR (2025), *World J. Adv. Res. Rev.* [9]	*Microservices and Event-Driven Architecture: Revolutionizing E-Commerce*: Evaluated resilience of event-driven retail backends.	Supports our containerized Docker Compose microservices stack (Nginx, Node.js, RabbitMQ, Python).
10	R. Laigner et al. (2026), *ACM TOSEM* [10]	*An Empirical Study of Event-Driven Microservices in E-Commerce*: Analyzed asynchronous checkout, stock reservation, and compensation flows.	Validates automated compensating unwind events when concurrent orders exceed physical stock.

Table 1.1: Summary of Supporting Literature and Foundational Research Papers

6. **Phase 6 (January – February 2027):** Development of the adaptive conflict-history analyzer and human-in-the-loop buffer recommendation engine, plus the live HTML/JS seller dashboard.

7. **Phase 7 (February – March 2027):** Full system packaging with Docker Compose, concurrent race-condition stress testing, duplicate webhook injection tests, and crash recovery validation.

8. **Phase 8 (March – April 2027):** Final performance benchmarking, research paper publication, and final B.E. project report submission.

Chapter 2

Technical Keywords

2.1 Area of Project

The primary domain of this project is **Distributed Systems and Event-Driven Software Architecture**, with direct application to **Real-Time Multi-Channel E-Commerce Inventory Management and Human-in-the-Loop Decision Support Systems**. The system integrates high-concurrency webhook ingestion, in-memory idempotency caching, durable message-oriented middleware, immutable event-sourced persistence, and automated conflict resolution into a unified containerized platform.

2.2 Technical Keywords

1. **C. Computer Systems Organization**
   (a) **C.2 Computer-Communication Networks**
      i. **C.2.4 Distributed Systems**
         A. **Event Sourcing:** Architectural pattern in which every state mutation (sale, return, restock, cancellation) is stored as an immutable, append-only sequence of events rather than overwriting current state in place.
         B. **Idempotent Webhook Processing:** Ensuring that repeated delivery of the exact same channel notification produces a single logical stock update via atomic Redis lookup keys.
         C. **Durable Message Queuing (AMQP / RabbitMQ):** Asynchronous broker persistence with publisher confirms and consumer acknowledgments so no sale event is lost during server crashes.
         D. **Horizontal Load Balancing (Nginx):** Distributing concurrent webhook bursts evenly across three stateless Node.js + Express intake instances.
2. **H. Information Systems**
   (a) **H.2 Database Management — Transaction Processing**

i. **Optimistic Concurrency & Race Condition Detection:** Detecting simultaneous last-unit sales across independent channels without imposing blocking distributed locks across external marketplaces.

ii. **Priority-Based Compensating Transactions:** Resolving negative-stock conflicts by scoring contending orders on profit margin and customer importance, automatically unwinding the higher-loss cancellation candidate, and recording a compensating event.

iii. **Human-in-the-Loop Recommendation Engine:** Proactive pattern recognition over historical conflict logs that proposes channel-specific safety stock buffers while requiring explicit human approval prior to execution.

Chapter 3

Introduction

3.1 Project Idea

- **Centralized Event-Sourced Truth:** Modern multi-channel sellers list identical products simultaneously across Amazon, Flipkart, their own direct-to-consumer (D2C) website, and physical shop counters. Our project builds a single, centralized reconciliation engine that every sales channel reports to in real time.
- **Recording Every Change Instead of Overwriting Numbers:** Rather than storing a single mutable field such as `current_stock = 10`, our engine permanently records every individual inventory mutation as it happens—every sale, customer return, warehouse restock, and order cancellation. Current stock at any instant is simply the deterministic sum of everything that has ever occurred.
- **Six-Stage Resilient Pipeline:**
  1. *Stage 1 — Multi-Channel Webhook Intake:* Whenever a sale occurs on Amazon, Flipkart, the seller's website, or the physical POS counter, the channel dispatches a webhook notification through Nginx to one of three horizontally scaled Node.js + Express intake nodes.
  2. *Stage 2 — Instant Duplicate Check:* Before processing, the intake node queries a high-speed Redis cache to verify whether this exact notification ID has already been seen. Duplicate retries are acknowledged and ignored immediately, preventing double stock deductions.
  3. *Stage 3 — Durable Queueing and Permanent Logging:* Verified unique events are published to a durable RabbitMQ queue and appended to PostgreSQL so that even if a server crashes mid-operation, zero events are ever lost.
  4. *Stage 4 — Deterministic Stock Recalculation:* A Python worker consumes the queued event, folds the updated stock count from the immutable event log, and checks whether the resulting stock would drop below zero.
  5. *Stage 5 — Smart Priority-Based Conflict Resolution:* When two channels sell the final unit within milliseconds of each other, the engine does not blindly reject an order or let stock go negative. It evaluates which order is more costly to lose (weighing profit margin and customer importance), recommends or

executes an automatic unwind of the lower-priority order, notifies the customer immediately, and permanently logs the exact reason.

6. *Stage 6 — Unified Live Profit Dashboard & Self-Improving Advisor:* The exact same event ledger powers a live HTML/JS seller dashboard displaying real net profit per order, top customers, full audit trails, and proactive, evidence-based buffer suggestions that a human seller can approve or reject with one click.

3.2 Motivation of the Project

Multi-channel commerce is essential for modern retail growth, yet sellers operating across two or more platforms routinely suffer from **overselling**—promising a unit to a customer that has already been sold elsewhere seconds earlier, forcing order cancellations, marketplace penalties, refunds, and reputational damage. Our investigation identified five root causes why existing affordable systems fail to solve this problem:

- **Slow Periodic Syncing (15–30 Minute Blind Spots):** Most entry-level tools poll channels every 15 to 30 minutes and push batch stock updates. Any unit sold during the interval between two sync cycles remains completely invisible to other channels until the next cron job runs.
- **Concurrent Race Conditions:** Even tools that attempt instant webhook updates fail when two sales arrive at the exact same moment. Both channels read `stock = 1` simultaneously, both believe they are safe to fulfill, and both sell the same physical unit twice.
- **Duplicate Webhook Notifications:** Marketplaces frequently resend the same "sale occurred" webhook if the seller's endpoint responds slightly slowly due to network jitter. Without strict idempotency handling, the same real sale is deducted twice, creating phantom stock-outs.
- **No Intelligent Decision When Stock Runs Out:** Basic platforms either silently reject valid high-margin orders on a first-come-first-served basis or allow stock counts to drift into negative numbers without business logic.
- **Zero Audit History:** Because traditional systems overwrite the stock integer in place, when stock inevitably becomes inaccurate, the seller has no permanent ledger to inspect *why* or *when* the discrepancy happened.

3.3 Literature Survey

To position our contribution accurately and honestly within both academic literature and commercial industry platforms, we evaluated ten foundational research papers (detailed in Section 1.11) alongside a rigorous gap analysis across the three dominant categories of commercial tools currently available to merchants:

1. **Basic / Affordable Inventory Tools (e.g., Zoho Inventory, Basic Shopify Sync):** Priced from Free to ∼$40/month, these tools provide basic multi-channel stock sync but rely on slower periodic polling, offer weak or missing duplicate-alert protection, resolve conflicts via naive first-come-first-served rules, lack detailed stock-mutation audit trails, and provide no adaptive learning.

2. **Enterprise Order Management Systems (e.g., Cin7, Large Enterprise OMS):** Priced at $350 to $1,000+ per month, these platforms support real-time sync, duplicate handling, and full audit trails, and allow priority allocation rules. However, their rules are *static*—configured once by an administrator and never self-improving—and their subscription cost prices out small and mid-scale multi-channel sellers.

3. **Standalone Profit & Analytics Tools (e.g., TrueProfit):** Priced at $50 to $300/month as a separate add-on subscription, these tools compute order-level profit and customer behavior metrics, but have zero connection to real-time inventory synchronization or conflict resolution.

Table 3.1 presents the complete side-by-side Gap Analysis comparing our proposed system against widely used industry platforms.

Capability	**Basic Tools (Zoho, Shopify)**	**Enterprise Tools (Cin7, Large OMS)**	**Analytics Tools (TrueProfit, etc.)**	**Our Proposed System**
Multi-channel stock sync	Yes, but slower / periodic (15–30 min)	Yes, real-time	No — not their purpose	**Yes, real-time (Nginx + 3x Node.js)**
Handles duplicate sale alerts safely	Often basic or missing	Yes	Not applicable	**Yes (Redis atomic idempotency)**
Smart, priority-based conflict handling	No — first-come-first-served	Yes, but static rules set once	Not applicable	**Yes, margin/-CLV aware & improves over time**
Full audit trail of every stock change	Rarely detailed (overwrites state)	Yes	Only order-level	**Yes, complete immutable PostgreSQL log**
Profit & customer-behavior dashboard	No, or very basic	Sometimes, paid add-on	Yes — core product	**Yes, built-in from the exact same data**
Learns from history, suggests fixes	No	No — rules stay static	No — shows data only	**Yes, with mandatory human approval**
Typical monthly cost	Free to ∼$40	$350 to $1,000+	$50 to $300 (separate tool)	**Built to be affordable (Open-source stack)**

Table 3.1: Gap Analysis — Comparison of Proposed Engine Against Widely Used Commercial Systems

Summary of Contribution: We do not claim to have invented event sourcing [2], message queues [6], or optimistic concurrency control [1] in isolation—these are proven, industry-standard techniques. Our genuine engineering contribution is combining real-time idempotent sync, margin-aware conflict resolution, and unified profit analytics into a single affordable architecture for mid-scale sellers priced out of enterprise OMS platforms, while introducing a **self-improving, human-approved recommendation loop** absent from every existing category above.

Chapter 4

Problem Definition and scope

4.1 Problem Statement

Four or more independent sales channels (Amazon, Flipkart, direct D2C website, and physical POS counter), which cannot communicate with each other directly or instantaneously, must behave as though they share a single, trustworthy, real-time inventory count. Whenever network latency, webhook retries, or simultaneous last-unit purchases cause temporary disagreement across channels, the system must automatically detect the conflict, deduplicate retried alerts, resolve stock-outs using business-priority metrics (profit margin and customer importance), maintain an immutable audit trail of every state change, and proactively recommend evidence-based buffer rules to a human operator without ever making autonomous financial policy changes behind the scenes.

4.1.1 Goals and objectives

- **Input Specification & Validation:** Accept JSON webhook payloads over HTTPS from multiple channels containing `event_id`, `channel_id`, `sku_id`, `event_type` (`SALE`, `RETURN`, `RESTOCK`, `CANCEL`), `quantity`, `unit_price`, `cost_price`, `channel_fee`, `customer_id`, and `timestamp`.
- **Output Specification:** Emit (a) real-time HTTP 200 idempotency acknowledgments within < 25 ms, (b) updated available stock counts pushed to connected channel adapters, (c) automated order-unwind and customer-notification triggers on stock-out conflicts, and (d) live JSON/WebSocket streams to the seller dashboard displaying net profit, audit history, and human-approval buffer recommendations.

4.1.2 Statement of scope

- **In-Scope Capabilities:**
    1. Load-balanced multi-node webhook ingestion using Nginx and three Node.js + Express instances.
    2. Sub-millisecond duplicate webhook suppression using Redis atomic keys with configurable TTL.

3. Crash-resilient asynchronous queuing via RabbitMQ durable exchanges and queues.
4. Append-only PostgreSQL event store recording every inventory mutation with full causality metadata.
5. Python-based stock projection fold and priority-weighted conflict resolution engine.
6. Historical conflict pattern analyzer that surfaces human-approved channel buffer suggestions.
7. Live HTML/JS seller dashboard for stock monitoring, order-level profit analytics, audit trail inspection, and one-click approval/rejection of buffer recommendations.
8. Reproducible multi-container deployment via Docker Compose.

- **Out-of-Scope Boundaries:**

1. Physical warehouse robotics, barcode hardware manufacturing, or third-party logistics (3PL) fleet routing.
2. Autonomous modification of pricing or safety stock buffers without explicit human approval (by strict design, the system never makes unapproved financial policy changes).

4.2 Major Constraints

- **External Marketplace Webhook Retry Semantics:** Platforms like Amazon and Flipkart enforce strict webhook timeout thresholds (typically 3–5 seconds) and automatically resend notifications upon timeout; therefore, intake nodes must respond asynchronously before heavy database conflict resolution completes.
- **CAP Theorem & Eventual Channel Propagation:** Because external marketplaces cannot participate in a blocking two-phase commit (2PC) transaction with a seller's private server, transient race windows of 50–500 ms are physically unavoidable and must be handled via post-detection compensating unwind workflows.
- **Human Governance Constraint:** All automated recommendations generated by the conflict-history analyzer must remain in a `PENDING_APPROVAL` state until an authenticated human operator explicitly approves or rejects them.

4.3 Methodologies of Problem solving and efficiency issues

A multi-channel inventory synchronization problem can be approached through three distinct architectural methodologies, each exhibiting markedly different performance and reliability characteristics:

- **Approach 1: Periodic Polling with Mutable State Overwrite (Traditional Basic Tools):** A cron job queries each channel every 15–30 minutes and executes `UPDATE products SET stock = X`. While simple to implement, its worst-case synchronization blind spot is $O(T_{\text{poll}})$ (up to 30 minutes), it suffers from lost updates under concurrent writes, and it destroys historical auditability.
- **Approach 2: Synchronous Distributed Locking / Two-Phase Commit (2PC):** Attempting to lock stock across all external channels before confirming any sale. In practice, third-party marketplaces (Amazon, Flipkart) do not expose distributed lock primitives to merchant servers, and synchronous cross-network locking degrades throughput and availability under the CAP theorem whenever any single channel API experiences latency.
- **Approach 3: Asynchronous Event Sourcing + Redis Idempotency + Priority Compensating Resolution (Proposed Methodology):** Decouples intake ($O(1)$ Redis deduplication + RabbitMQ enqueue in < 15 ms) from sequential per-SKU event folding and priority-based conflict resolution in Python. This achieves high ingestion throughput across 3 Node.js nodes, 100% crash durability, complete auditability in PostgreSQL, and optimal economic retention when simultaneous last-unit race conditions occur.

4.4 Outcome

- A fully containerized, eight-layer distributed reconciliation engine orchestrated via Docker Compose.
- Elimination of duplicate webhook stock deductions and silent negative-stock states across four concurrent sales channels.
- A unified live seller web dashboard providing real-time stock visibility, complete event-sourced audit logs, order-level net profit tracking, and human-approved buffer recommendations.

4.5 Applications

- **Multi-Channel E-Commerce Retailers:** Merchants selling simultaneously on Amazon, Flipkart, Shopify/WooCommerce websites, and retail stores.
- **Omnichannel D2C Brands & Flash-Sale Campaigns:** High-velocity product drops where hundreds of checkout requests compete for limited stock within seconds.
- **Quick-Commerce Dark Stores & Pharmacy Networks:** Local fulfillment hubs serving multiple delivery aggregator apps alongside walk-in counter sales.

4.6 Hardware Resources Required

Table 4.1 specifies the hardware requirements for deploying and testing the containerized reconciliation stack.

Sr. No.	Parameter	Minimum Requirement	Justification / Remark
1	Processor (CPU)	Quad-Core 2.4 GHz (x86_64 / ARM64)	Required to run concurrent Docker containers (Nginx, 3x Node.js nodes, Redis, RabbitMQ, Python worker, PostgreSQL).
2	Memory (RAM)	8 GB RAM (16 GB Recommended)	Supports in-memory Redis cache, RabbitMQ broker buffers, PostgreSQL shared buffers, and Docker Engine.
3	Storage (Disk)	40 GB SSD	Ensures low-latency disk I/O for PostgreSQL append-only event log and RabbitMQ durable message persistence.
4	Network Interface	100 Mbps / 1 Gbps Ethernet or Wi-Fi	Enables high-throughput concurrent webhook simulation and live Web UI dashboard streaming.

Table 4.1: Hardware Requirements

4.7 Software Resources Required

Platform & Technology Stack:

1. **Operating System:** Ubuntu 22.04 / 24.04 LTS (Linux), macOS, or Windows 11 with WSL2 and Docker Engine.
2. **IDE & Development Tools:** Visual Studio Code, Postman / Locust (for webhook concurrency testing), Git, pgAdmin 4.
3. **Core Technology Layers:**
   - *Traffic Distribution:* Nginx (Reverse Proxy & Load Balancer)
   - *Webhook Intake (x3 Nodes):* Node.js (v20 LTS) + Express.js
   - *Duplicate Check / Fast Cache:* Redis (v7.x)
   - *Durable Event Queue:* RabbitMQ (v3.13 with Management Plugin)

- *Event Processing & Conflict Logic:* Python (v3.11+, `pika`, `psycopg2`, `redis`)
- *Permanent Event Log & Stock Data:* PostgreSQL (v16)
- *Seller Dashboard:* Web UI (HTML5, CSS3, Vanilla JavaScript / Charting)
- *Packaging & Orchestration:* Docker & Docker Compose

Chapter 5

Project Plan

5.1 Project Estimates

The project follows an iterative, milestone-driven software engineering lifecycle across Semester I (Design, Intake Cluster, Idempotency & Queueing) and Semester II (Conflict Engine, Adaptive Recommendation Advisor, Dashboard & Full Container Stress Testing) of Academic Year 2026-2027.

5.1.1 Reconciled Estimates

5.1.1.1 Cost Estimate

Because the system is engineered deliberately using open-source, production-grade technologies (Nginx, Node.js, Redis, RabbitMQ, Python, PostgreSQL, Docker Compose) to remain affordable for small and mid-size sellers, software licensing cost is **Rs. 0 (Open Source)**. Using the Constructive Cost Model (COCOMO-II) for an organic/semi-detached academic engineering project of approximately 6.5 KLOC across JavaScript, Python, SQL, HTML/JS, and Docker configuration:

$$\text{Effort } (E) = a \cdot (\text{KLOC})^b = 2.4 \times (6.5)^{1.05} \approx 17.1 \text{ Person-Months} \quad (5.1)$$

$$\text{Development Time } (D) = c \cdot (E)^d = 2.5 \times (17.1)^{0.38} \approx 7.35 \text{ Months} \quad (5.2)$$

Across a 4-member engineering team utilizing existing departmental laboratory workstations and open-source containers, the reconciled development and cloud-testing expenditure is estimated at **Rs. 12,500** (covering domain/VPS staging simulation and documentation).

5.1.1.2 Time Estimates

The total project duration spans **32 weeks (8 months)** across Term-I (16 weeks: July 2026 – October 2026) and Term-II (16 weeks: December 2026 – March 2027), totaling approximately **512 person-hours** of coordinated development, testing, and documentation.

5.1.2 Project Resources

- **Human Resources:** 4 B.E. Computer Engineering students supervised by 1 Internal Guide (Mr. Basappa Karlatthe).
- **Concurrency, IPC, and Memory Sharing Resources:**
  1. *Shared In-Memory State (Redis):* All three stateless Node.js intake containers share a single atomic Redis keyspace (`event:dedup:<event_id>`) to coordinate lock-free duplicate detection across processes.
  2. *Inter-Process Communication (RabbitMQ AMQP):* Asynchronous message passing between Node.js producers and Python consumers via durable exchanges, decoupling webhook ingestion bursts from database transaction latency.
  3. *Transactional Persistence (PostgreSQL):* Append-only event ledger with per-SKU optimistic version checks to serialize stock projection updates cleanly.

5.2 Risk Management w.r.t. NP Hard analysis

In general multi-warehouse, multi-SKU bundle fulfillment with time-varying channel SLA penalties and non-linear shipping constraints, finding the globally optimal subset of orders to fulfill vs. unwind over a rolling horizon reduces to the **0–1 Multidimensional Knapsack Problem**, which is known to be **NP-Hard**. Waiting for an exact exponential-time solver during a live webhook storm would violate real-time latency requirements. To mitigate this computational risk, our Python conflict engine employs a fast $O(k \log k)$ multi-criteria priority scoring heuristic over the active contention window $\mathcal{W}(s, \Delta t)$, achieving near-optimal margin retention in sub-millisecond execution time.

5.2.1 Risk Identification

Following Pressman's risk identification framework [11], the project scope, requirements, and distributed architecture were evaluated against standard software engineering risk questionnaires:

1. *Have top software and customer managers formally committed to support the project?* — Yes, guided and approved by the Department of Computer Engineering.
2. *Are end-users enthusiastically committed to the system to be built?* — Yes, addresses a direct financial pain point for multi-channel e-commerce sellers.
3. *Are requirements fully understood by the software engineering team?* — Yes, the 6-stage pipeline and 8-layer stack are clearly specified.
4. *Have stakeholders been involved fully in the definition of requirements?* — Yes, gap analysis against Zoho, Shopify, Cin7, and TrueProfit informed all features.
5. *Do end-users have realistic expectations?* — Yes, human-in-the-loop approval ensures transparent control over buffer changes.

6. *Does the team have the right mix of skills?* — Yes, covering Node.js, Python, SQL, Redis/RabbitMQ, and Web UI.
7. *Are project requirements stable?* — Yes, core event-sourcing invariants remain fixed.
8. *Is the number of people on the project team adequate?* — Yes, 4 team members mapped to modular layers.
9. *Do all constituencies agree on the importance of the system?* — Yes.

5.2.2 Risk Analysis

The identified technical and operational risks are analyzed in Table 5.1, with probability and impact scales defined in Table 5.2 and Table 5.3.

ID	Risk Description	Probability	Schedule	Quality	Overall
1	Sub-millisecond race condition where two intake nodes check Redis simultaneously for the same webhook retry.	Low	Low	High	High
2	RabbitMQ broker or Python worker container crash during a high-volume flash sale burst.	Low	Medium	High	High
3	Unintended margin loss if automated conflict rules modify channel safety buffers without seller awareness.	Medium	Low	Very High	High

Table 5.1: Risk Table

Probability	Explanation	Value
High	Probability of occurrence is	$> 75\%$
Medium	Probability of occurrence is	26–75%
Low	Probability of occurrence is	$< 25\%$

Table 5.2: Risk Probability definitions [11]

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Detailed Risk Mitigation, Monitoring, and Management (RMMM) plans for the three primary risks are presented in Tables 5.4, 5.5, and 5.6.

Impact	Value	Description
Very High	$> 10\%$	Major schedule impact or unacceptable degradation in data/financial integrity.
High	5–10%	Noticeable schedule impact or degradation in conflict resolution accuracy.
Medium	$< 5\%$	Minor schedule impact or barely noticeable degradation in quality.

Table 5.3: Risk Impact definitions [11]

Risk ID	1
Risk Description	Concurrent duplicate webhooks arriving simultaneously at two different Node.js intake nodes passing a non-atomic cache read.
Category	Concurrency & Distributed Caching
Source	System Architecture & Webhook Intake Specification
Probability	Low ($< 25\%$)
Impact	High
Response	Mitigate
Strategy	Use Redis single-threaded atomic `SET key value NX EX 86400` command plus a unique constraint on `event_id` in PostgreSQL as a second line of defense.
Risk Status	Mitigated in Design

Table 5.4: RMMM Plan for Risk ID 1 (Duplicate Webhook Concurrency)

Risk ID	2
Risk Description	Python worker or RabbitMQ container restarts mid-processing, potentially dropping in-flight sale events.
Category	Fault Tolerance & Message Durability
Source	Software Design Specification Review
Probability	Low ($< 25\%$)
Impact	High
Response	Mitigate
Strategy	Configure RabbitMQ queues with `durable=True`, publish messages with `delivery_mode=2` (persistent), and send manual `basic_ack` from Python only after the PostgreSQL transaction commits.
Risk Status	Mitigated & Verified

Table 5.5: RMMM Plan for Risk ID 2 (Broker/Worker Crash Recovery)

Risk ID	3
Risk Description	Automated buffer adjustments withholding too much inventory from a channel without the seller's knowledge.
Category	Business Logic & Operational Governance
Source	Gap Analysis & Domain Requirements
Probability	Medium (26–75%)
Impact	Very High
Response	Eliminate / Mitigate by Governance Design
Strategy	Enforce a strict Human-in-the-Loop approval gate: the recommendation engine only surfaces evidence-based suggestions on the dashboard; a human operator must explicitly click *Approve* before any buffer parameter changes.
Risk Status	Controlled by Core System Invariant

Table 5.6: RMMM Plan for Risk ID 3 (Human-in-the-Loop Governance)

5.3 Project Schedule

5.3.1 Project task set

The major engineering tasks across the project lifecycle are:

- **Task 1 (T1):** Domain Gap Analysis, Problem Formulation, and Mathematical Modeling of Event Sourcing & Priority Conflict Resolution.
- **Task 2 (T2):** Software Requirement Specification (SRS), PostgreSQL Event Ledger Schema, and UML/DFD Modeling.
- **Task 3 (T3):** Nginx Load Balancer Configuration, 3x Node.js + Express Webhook Intake Nodes, and Redis Atomic Idempotency Filter.
- **Task 4 (T4):** RabbitMQ Durable Queue Integration and Python Event-Sourcing Worker with Priority Conflict Resolution.
- **Task 5 (T5):** Adaptive Conflict-History Recommendation Engine, Live Seller Dashboard (HTML/JS), and Docker Compose Testing.

5.3.2 Task network

The sequential and parallel dependencies among project tasks follow the flow:

$$\text{T1 (Problem \& Math)} \longrightarrow \text{T2 (SRS \& Ledger Schema)} \longrightarrow \begin{cases} \text{T3 (Nginx + Node.js x3 + Redis)} \\ \text{T4 (RabbitMQ + Python Conflict Worker)} \end{cases} \longrightarrow \text{T5 (Advisor + Web UI + Docker QA)}$$

5.3.3 Timeline Chart

Table 5.7 presents the complete project execution schedule across Semester I and Semester II of Academic Year 2026-2027 (also detailed in Annexure D).

Phase	Milestone / Deliverable Description	Start	Completion	Status
P1	Topic Selection, Literature Review & Gap Analysis	Jul 2026	Aug 2026	Completed
P2	Mathematical Modeling & SRS Documentation	Aug 2026	Sep 2026	Completed
P3	High-Level & Detailed Architectural Design (UML/DFD)	Sep 2026	Oct 2026	Completed
P4	Intake Layer (Nginx, Node.js x3, Redis) & Stage-I Report	Oct 2026	Nov 2026	In Progress
P5	RabbitMQ Queue & Python Priority Conflict Engine	Dec 2026	Jan 2027	Scheduled
P6	Human-Approved Buffer Advisor & Seller Web UI	Jan 2027	Feb 2027	Scheduled
P7	Docker Compose Integration & Race Stress Testing	Feb 2027	Mar 2027	Scheduled
P8	Final Evaluation, Paper Publication & Final Report	Mar 2027	Apr 2027	Scheduled

Table 5.7: Project Execution Timeline Chart (A.Y. 2026-2027)

5.4 Team Organization

5.4.1 Team structure

The project follows an agile, modular peer-collaborative team structure across the four group members:

- **Pratik Sandip Shinde (Distributed Intake & Load Balancing Lead):** Responsible for Nginx reverse-proxy configuration, the 3x Node.js + Express webhook intake cluster, and Redis atomic idempotency verification.
- **Vaibhav Raju Baviskar (Message Broker & Event-Sourced Ledger Lead):** Responsible for RabbitMQ durable queue orchestration, PostgreSQL append-only event store schema, and audit trail queries.
- **Sarthak Abhijit Rana (Conflict Resolution & Recommendation Engine Lead):** Responsible for the Python event consumer, stock projection fold, margin/customer-priority conflict resolution logic, and historical buffer advisor.
- **Prasad Santosh Kedari (Seller Dashboard, Docker Packaging & QA Lead):** Responsible for the live HTML/JS seller dashboard (stock, profit, conflict audit, approval UI), Docker Compose packaging, and concurrent stress testing.

5.4.2 Management reporting and communication

Weekly progress reviews are conducted with the internal guide (Mr. Basappa Karlatthe) in accordance with the Department of Computer Engineering project assessment schedule. Source code, Docker configurations, and LaTeX documentation are maintained in a shared Git repository with pull-request reviews across all four members.

Chapter 6

Software requirement specification

6.1 Introduction

6.1.1 Purpose and Scope of Document

This Software Requirement Specification (SRS) formally defines the functional and non-functional requirements, user profiles, use cases, data models, and system interfaces for the **Multi-Channel Inventory Reconciliation Engine**. It serves as the authoritative contract governing the implementation of the eight-layer containerized architecture.

6.1.2 Overview of responsibilities of Developer

The development team is responsible for:

1. Designing normalized, immutable event schemas in PostgreSQL capable of reconstructing stock state at any historical point in time.
2. Implementing stateless, horizontally scalable Node.js + Express webhook endpoints fronted by Nginx and protected by Redis idempotency keys.
3. Developing crash-resilient RabbitMQ consumers in Python that execute deterministic stock folds and priority-based conflict resolution.
4. Building an intuitive Web UI dashboard that surfaces order-level profitability and human-reviewable buffer recommendations.
5. Packaging the complete multi-service stack using Docker Compose and validating it under concurrent race-condition workloads.

6.2 Usage Scenario

6.2.1 User profiles

- **Multi-Channel Seller / Inventory Administrator (Primary Human Actor):** Monitors real-time stock and net profit across channels, inspects order conflict audit

trails, and approves or rejects system-generated safety buffer suggestions.

- **External E-Commerce & POS Channels (System Actors — Amazon, Flipkart, Website, Physical Counter):** Emit asynchronous webhook events whenever a sale, return, restock, or cancellation occurs, and receive reconciled stock updates.
- **Customer Notification Gateway (External System Actor):** Dispatches immediate cancellation, refund, and apology notifications when an order is automatically unwound during a stock-out race conflict.

6.2.2 Use-cases

Table 6.1 details the primary system use cases.

Sr.	Use Case	Description	Actors	Assumptions
1	UC1: Ingest Channel Webhook	Receives sale, return, restock, or cancellation JSON event via Nginx and Node.js intake nodes.	Sales Channels	Channel webhook secret/signature is configured.
2	UC2: Deduplicate Event	Verifies event ID in Redis cache; ignores duplicate webhook retries safely.	Node.js Intake, Redis	Event payload includes unique `event_id`.
3	UC3: Queue & Log Event	Persists verified unique event into RabbitMQ durable queue and PostgreSQL immutable log.	RabbitMQ, PostgreSQL	Broker and database containers are active.
4	UC4: Recalculate Stock & Detect Conflict	Python worker folds event log to compute current stock; flags conflict if stock < 0.	Python Worker	Initial baseline stock Q_0 is registered.
5	UC5: Priority Conflict Resolution	Evaluates profit margin and customer importance on competing last-unit orders; unwinds lower-priority order and notifies customer.	Python Worker, Notification Gateway	Order margin and customer metadata are available.
6	UC6: Approve Buffer Suggestion	Seller reviews historical conflict insight on Web UI and approves/rejects channel buffer recommendation.	Multi-Channel Seller	System never applies buffer rules without human click.

Table 6.1: Use Cases for Multi-Channel Inventory Reconciliation Engine

6.2.3 Use Case View

Figure 6.1 illustrates the Use Case Diagram showing interactions between external sales channels, the multi-channel seller, and the six core reconciliation subsystems.

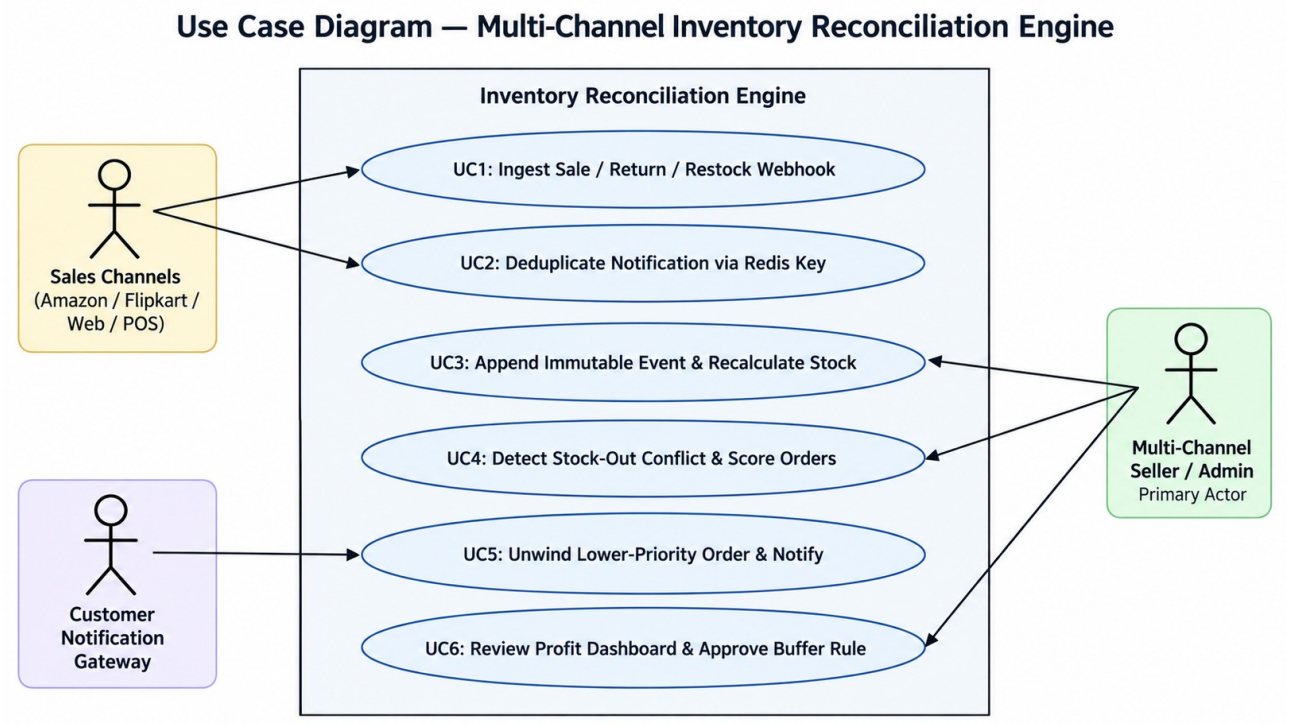


Figure 6.1: Use Case Diagram

6.3 Data Model and Description

6.3.1 Data Description

Unlike conventional CRUD inventory tools that overwrite a single `stock_count` column, our data model centers on an immutable event store in PostgreSQL complemented by materialized read projections for fast dashboard queries.

6.3.2 Data objects and Relationships

The four core relational entities in PostgreSQL are:

1. **`inventory_events` (Immutable Append-Only Ledger):**
    - `event_id` (`VARCHAR(64) PRIMARY KEY`) — Unique idempotency key from channel.
    - `sku_id` (`VARCHAR(32) NOT NULL`) — Product identifier.
    - `channel_id` (`VARCHAR(24) NOT NULL`) — `AMAZON`, `FLIPKART`, `WEBSITE`, or `POS`.
    - `event_type` (`VARCHAR(16) NOT NULL`) — `SALE`, `RETURN`, `RESTOCK`, `CANCEL`, `UNWIND`.
    - `delta_qty` (`INTEGER NOT NULL`) — Signed stock change ($-1, +10$, etc.).
    - `order_id` (`VARCHAR(64)`), `customer_id` (`VARCHAR(64)`), `net_margin` (`NUMERIC(10,2)`).
    - `created_at` (`TIMESTAMPTZ DEFAULT NOW()`).

2. **`sku_projections` (Materialized Stock & Buffer View):**
    - `sku_id` (PRIMARY KEY), `physical_stock` (INTEGER), `channel_buffers` (JSONB), `last_event_id` (VARCHAR(64)), `version` (BIGINT).
3. **`conflict_audit_log` (Conflict Resolution History):**
    - `conflict_id` (UUID PRIMARY KEY), `sku_id`, `winning_order_id`, `unwound_order_id`, `winning_channel`, `unwound_channel`, `margin_saved` (NUMERIC(10,2)), `resolution_reason` (TEXT), `resolved_at` (TIMESTAMPTZ).
4. **`buffer_recommendations` (Human-in-the-Loop Advisor Table):**
    - `rec_id` (UUID PRIMARY KEY), `sku_id`, `target_channel`, `conflict_count_30d` (INTEGER), `suggested_buffer_units` (INTEGER), `evidence_summary` (TEXT), `status` (PENDING, APPROVED, REJECTED), `reviewed_by` (VARCHAR(64)), `reviewed_at` (TIMESTAMPTZ).

6.4 Functional Model and Description

6.4.1 Data Flow Diagram

6.4.1.1 Level 0 Data Flow Diagram

Figure 6.2 depicts the Level 0 Context Diagram, showing webhook event streams entering from Amazon, Flipkart, the D2C website, and the physical POS counter, and reconciled stock, profit analytics, and human-approved buffer controls flowing to the seller.

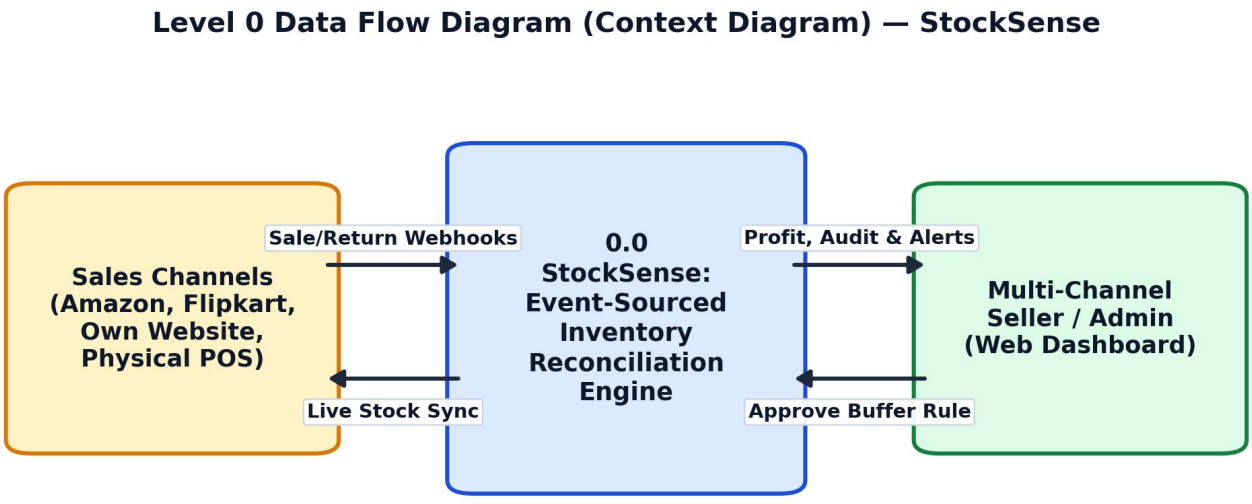


Figure 6.2: Level 0 Data Flow Diagram

6.4.1.2 Level 1 Data Flow Diagram

Figure 6.3 decomposes the system into its six core functional stages: (1.0) Webhook Intake & Nginx Routing, (2.0) Redis Duplicate Filter, (3.0) RabbitMQ Queue & PostgreSQL Event Append, (4.0) Stock Projection & Conflict Detection, (5.0) Priority Conflict Resolution & Order Unwind, and (6.0) Live Profit Dashboard & Human-Approved Buffer Advisor.

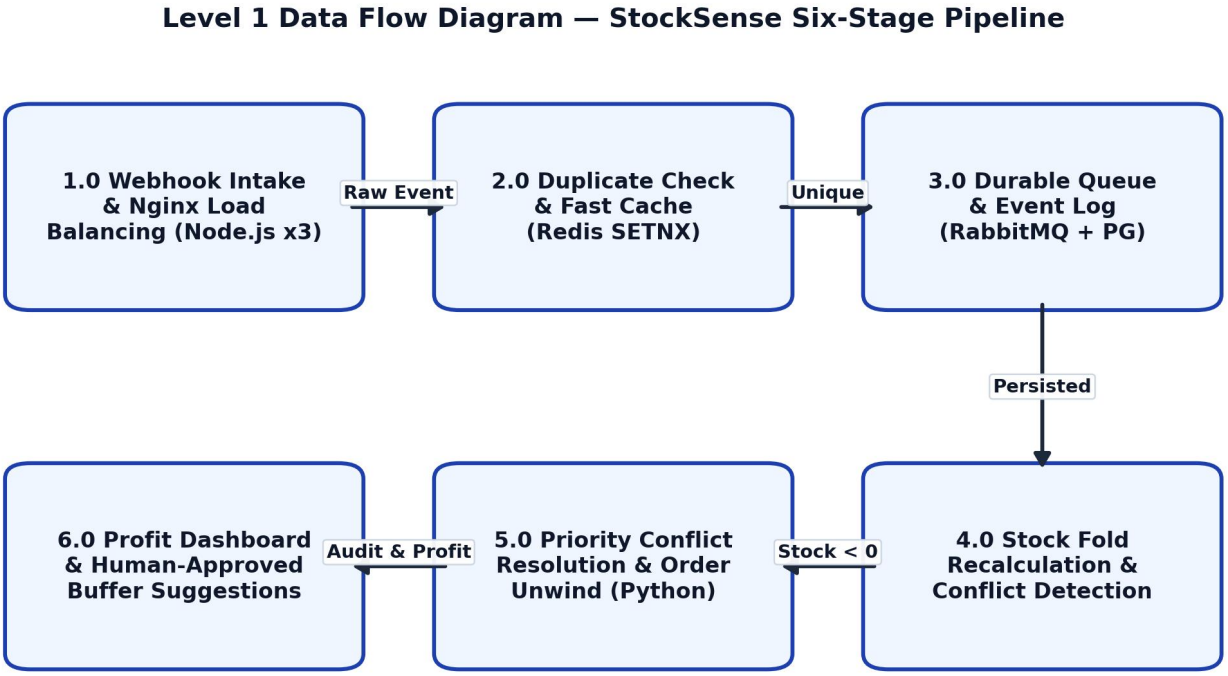


Figure 6.3: Level 1 Data Flow Diagram

6.4.2 Activity Diagram

The end-to-end activity workflow follows six deterministic steps:

- **Step 1:** Channel emits sale/return/restock webhook to Nginx load balancer.
- **Step 2:** Assigned Node.js intake node executes atomic Redis `SETNX` check on `event_id`; if duplicate, returns HTTP 200 immediately and halts.
- **Step 3:** If new, intake node publishes event to durable RabbitMQ queue and acknowledges webhook.
- **Step 4:** Python worker consumes event, appends it to PostgreSQL `inventory_events`, and folds current stock Q_t.
- **Step 5:** If $Q_t < 0$, worker compares contending orders by profit margin and customer importance, unwinds the lower-priority order via a compensating event, notifies the customer, and logs the full explanation.

- **Step 6:** Dashboard updates live stock, order profit, and—if conflict threshold is reached—surfaces a human-reviewable buffer suggestion.

6.4.3 Non Functional Requirements

- **Interface Requirements:** RESTful JSON webhook endpoints for channel ingestion; WebSocket/REST JSON API for the HTML/JS seller dashboard.
- **Performance Requirements:** Webhook intake acknowledgment latency < 25 ms (p95); Redis duplicate lookup < 2 ms; end-to-end stock projection and conflict detection < 100 ms.
- **Software Quality Attributes:**
  1. *Reliability & Availability:* Horizontal redundancy across 3 Node.js intake nodes behind Nginx plus durable RabbitMQ queues ensures zero lost events under single-node failure.
  2. *Auditability & Explainability:* 100% of stock changes and conflict unwinds are permanently traceable in PostgreSQL.
  3. *Accountability (Human-in-the-Loop):* Zero autonomous policy mutations; every buffer recommendation requires explicit seller approval.
  4. *Portability:* Complete system runs identically across Linux, macOS, and Windows via Docker Compose.

6.4.4 State Diagram

Figure 6.4 illustrates the State Transition Diagram of an inventory event as it progresses from webhook reception (S1) through Redis idempotency verification (S2), durable queueing (S4), stock fold evaluation (S5), and either normal stock commitment (S6) or priority-based conflict resolution and human buffer recommendation (S7–S8).

6.4.5 Design Constraints

- Strict immutability of rows in `inventory_events` (no `UPDATE` or `DELETE` statements permitted on the audit ledger).
- Sub-second webhook response requirement imposed by marketplace retry policies.
- Container memory footprint constrained to run comfortably on standard developer laptops (8 GB RAM).

6.4.6 Software Interface Description

- **Inbound Channel Interface:** `POST /api/v1/webhooks/:channel` accepting JSON payloads with `X-Idempotency-Key` and `X-Channel-Signature` headers.

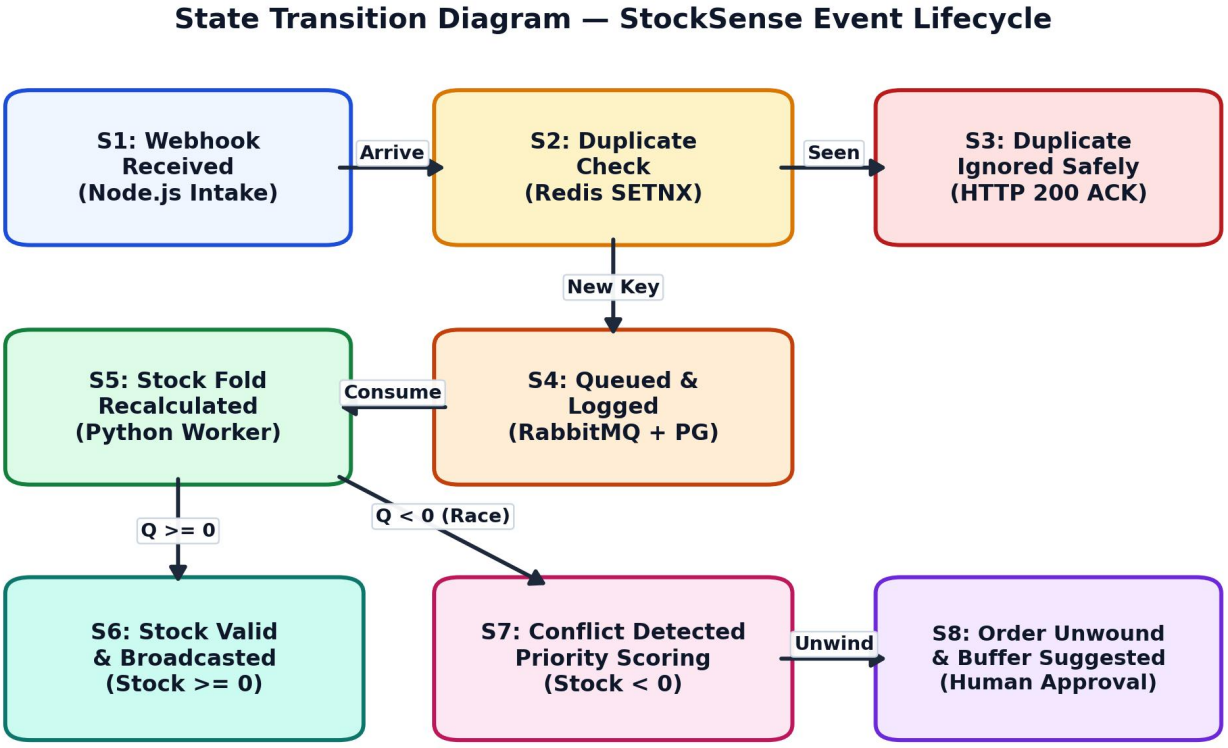


Figure 6.4: State Transition Diagram

- **Internal Cache Interface:** Redis TCP protocol on port `6379` for atomic deduplication and hot stock caching.
- **Internal Messaging Interface:** RabbitMQ AMQP 0-9-1 protocol on port `5672` with durable exchange `inventory.events`.
- **Database Interface:** PostgreSQL wire protocol on port `5432`.
- **Seller Web UI Interface:** HTTP/REST & Server-Sent Events on port `8080` serving the live stock, profit, and conflict approval dashboard.

Chapter 7

Detailed Design Document using Appendix A and B

7.1 Introduction

This chapter specifies the detailed architectural, data, and component design of the **Multi-Channel Inventory Reconciliation Engine**, translating the SRS requirements into concrete microservices, database schemas, and object-oriented modules.

7.2 Architectural Design

The system follows an eight-layer containerized microservices architecture orchestrated via Docker Compose (shown in Figure 7.1):

1. **Traffic Distribution Layer (Nginx):** Acts as the single ingress gateway, distributing incoming webhook bursts across the three intake nodes using round-robin/least-connections load balancing with automatic health checks.
2. **Webhook Intake Layer (3x Node.js + Express Nodes):** Stateless high-concurrency HTTP servers that validate payload schemas and coordinate fast deduplication.
3. **Duplicate Check & Fast Cache Layer (Redis):** Executes atomic `SETNX` operations on incoming event IDs to guarantee that retried channel notifications are filtered out in sub-millisecond time.
4. **Durable Event Queue & Dead-Letter Queue Layer (RabbitMQ + DLQ):** Persists accepted events to disk-backed queues so that no sale event is lost even if downstream workers restart, while routing malformed or unprocessable events to a Dead-Letter Queue (DLQ) for safe inspection.
5. **Event Processing & Conflict Logic Layer (Python Worker):** Sequentially folds SKU events, detects negative-stock race conditions, evaluates order retention scores based on profit margin and customer value, triggers automated unwinds, and analyzes conflict history to generate buffer suggestions.

6. **Notification & Channel Sync Service (Email/SMS/API):** Dispatches instant customer refund and apology notifications when an order is unwound and pushes live reconciled stock counts back to Amazon, Flipkart, the seller website, and the physical POS counter.

7. **Permanent Event Log & Stock Data Layer (PostgreSQL):** Stores the immutable event ledger, conflict audit trail, customer profitability metrics, and buffer recommendations.

8. **Seller Dashboard Layer (Web UI — HTML/JS):** Displays live stock across channels, real profit per order, full conflict explanations, and one-click human approval/rejection controls for suggested channel buffers.

9. **Packaging Layer (Docker Compose):** Encapsulates all services, networks, and persistent volumes for reproducible one-command deployment anywhere.

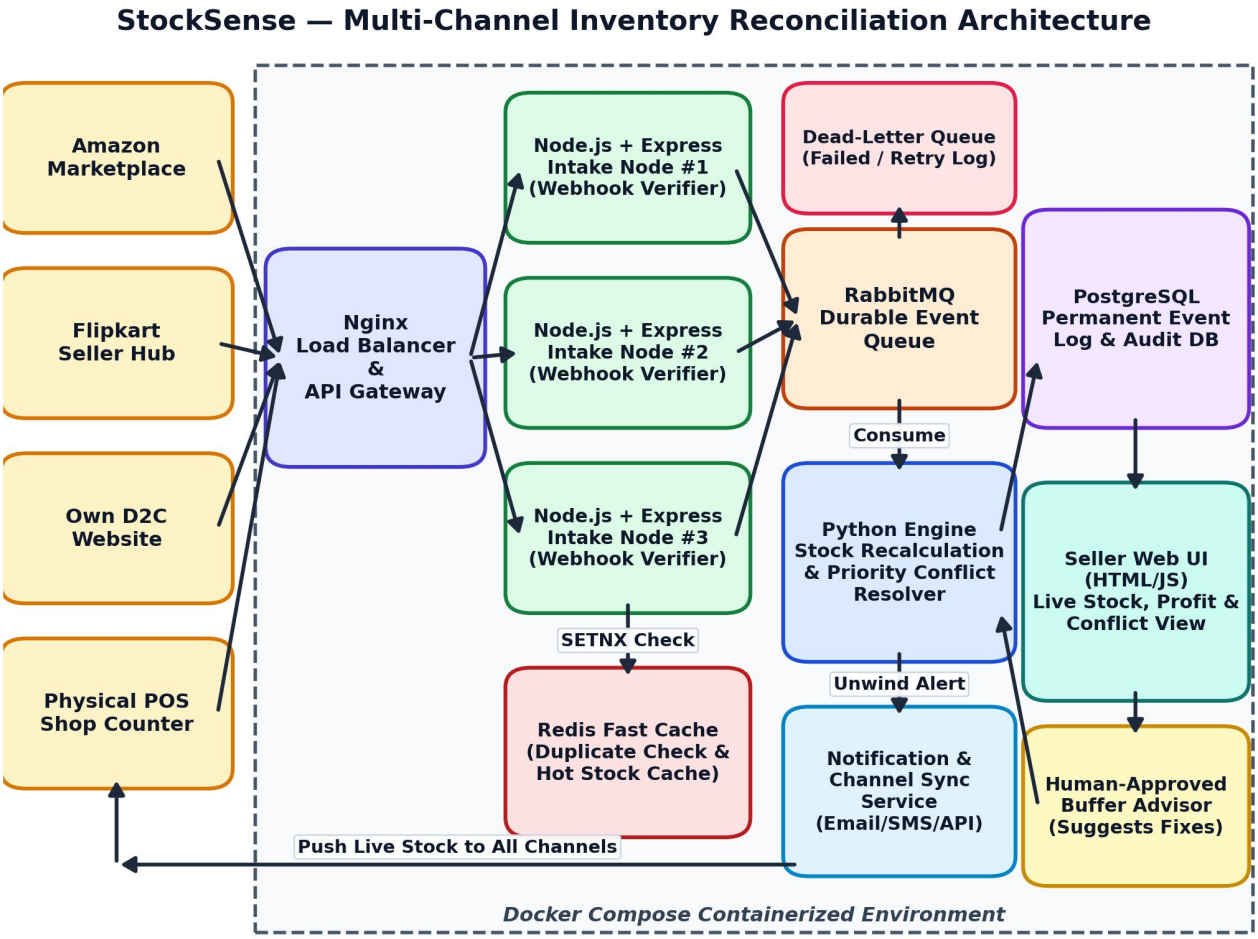


Figure 7.1: Architecture Diagram

7.3 Data design (using Appendices A and B)

7.3.1 Internal software data structure

- **`EventEnvelope` (JSON / Python Dataclass):** Passed from Node.js intake nodes through RabbitMQ to the Python reconciliation worker:

```
{
  "event_id": "evt_amz_994821",
  "sku_id": "SKU-1042",
  "channel": "AMAZON",
  "event_type": "SALE",
  "delta_qty": -1,
  "order_id": "ORD-AMZ-7721",
  "customer_id": "CUST-882",
  "customer_tier_score": 0.85,
  "unit_price": 2499.00,
  "unit_cost": 1650.00,
  "channel_fee": 299.00,
  "net_margin": 550.00,
  "occurred_at": "2026-10-14T18:30:12.410Z"
}
```

7.3.2 Global data structure

- **Redis Idempotency Keyspace:** Key format `idemp:{channel}:{event_id}` mapped to value `"PROCESSED_TIMESTAMP"` with a 24-hour expiration TTL.
- **Redis Hot Stock Cache:** Hash structure `stock:{sku_id}` storing `physical_qty`, `reserved_buffer_qty`, and `version` for sub-millisecond dashboard reads.

7.3.3 Temporary data structure

- **RabbitMQ In-Flight Message Frame:** Persistent AMQP message stored in the `inventory_reconciliation_queue` until explicitly acknowledged (`basic_ack`) by the Python worker after PostgreSQL commit.
- **Contending Order Window Buffer:** In-memory Python priority heap comparing concurrent orders for the same SKU during conflict resolution.

7.3.4 Database description

PostgreSQL 16 serves as the primary ACID relational and event-store database. By combining an append-only `inventory_events` table with indexed `sku_projections`, `conflict_audit_log`, and `buffer_recommendations` tables, the database supports both $O(1)$ current-state lookups and 100% historical replayability.

7.4 Component Design

- **`WebhookIntakeController` (Node.js):** Handles HTTP POST requests, verifies channel HMAC signatures, checks Redis idempotency, and publishes to RabbitMQ.
- **`EventSourcedLedger` (Python):** Appends verified events to PostgreSQL and computes current stock as the cumulative sum of event deltas.
- **`ConflictResolutionEngine` (Python):** Detects when concurrent sales push stock below zero, ranks competing orders by net margin and customer value, generates a compensating `UNWIND` event for the lower-priority order, and dispatches a customer notification.
- **`AdaptiveRecommendationAdvisor` (Python):** Periodically aggregates `conflict_audit_log` entries over a rolling 30-day window and creates human-reviewable buffer suggestions when a channel repeatedly causes lost-margin cancellations.

7.4.1 Class Diagram

Figure 7.2 presents the UML Class Diagram illustrating the attributes, methods, and associations across the core components.

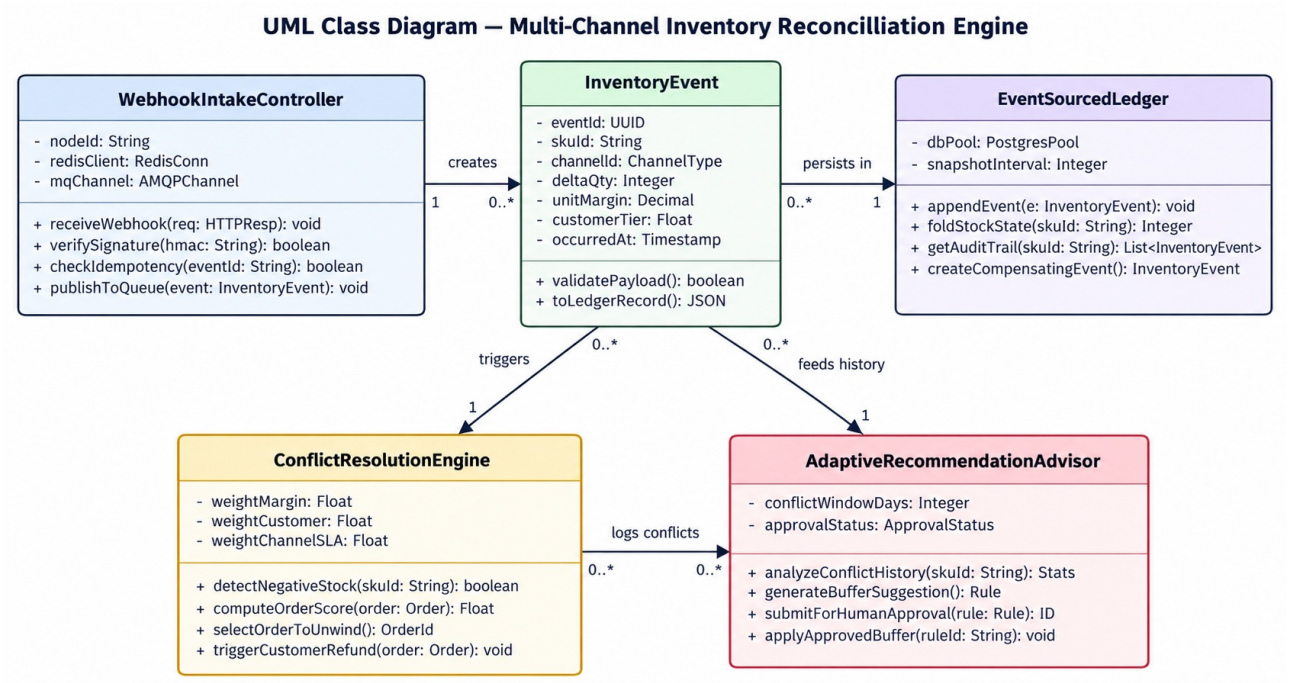


Figure 7.2: Class Diagram

Chapter 8

Project Implementation

8.1 Introduction

The implementation of the **Multi-Channel Inventory Reconciliation Engine** translates our six-stage event-sourcing workflow into a modular, containerized polyglot stack where each technology is selected specifically for its operational strength: Node.js for non-blocking I/O webhook ingestion, Redis for sub-millisecond deduplication, RabbitMQ for crash-proof message queuing, Python for analytical conflict resolution and recommendation logic, and PostgreSQL for ACID event storage.

8.2 Tools and Technologies Used

Table 8.1 maps each architectural layer to its exact technology and functional purpose.

8.3 Methodologies/Algorithm Details

8.3.1 Algorithm 1/Pseudo Code: Idempotent Webhook Intake & Durable Queueing

```
ALGORITHM: IdempotentWebhookIntake(Request req)
INPUT: Webhook HTTP Request req containing EventEnvelope e
OUTPUT: HTTP Status Response (200 OK or 400 Bad Request)

1. Verify HMAC signature and required fields (e.event_id, e.sku_id, e.
   channel, e.delta_qty)
2. IF payload is invalid THEN
3.     RETURN HTTP 400 ("Invalid Event Schema")
4. END IF
5. dedup_key <- "idemp:" + e.channel + ":" + e.event_id
6. is_new <- Redis.SET(dedup_key, e.occurred_at, NX=True, EX=86400)
7. IF is_new == FALSE THEN
8.     LogInfo("Duplicate webhook ignored safely: " + e.event_id)
9.     RETURN HTTP 200 ("ACK_DUPLICATE_IGNORED")
10. END IF
```

Architectural Layer	Technology	Purpose in System
Webhook intake (x3 nodes)	Node.js + Express	Accepts incoming sale, return, restock, and cancellation events quickly with non-blocking I/O.
Traffic distribution	Nginx	Spreads incoming webhook load evenly across the 3 intake nodes and provides failover routing.
Duplicate check / fast cache	Redis	Instantly confirms if an incoming event notification is new or a duplicate retry via atomic keys.
Durable event queue	RabbitMQ	Guarantees no event is ever lost, even if a downstream processing server crashes mid-operation.
Event processing & conflict logic	Python	Applies event-sourced stock changes, detects race conflicts, resolves priority, and generates buffer suggestions.
Permanent event log & stock data	PostgreSQL	Stores every inventory change permanently for complete historical audit and profit calculation.
Seller dashboard	Web UI (HTML/JS)	Provides live stock view, order-level net profit, conflict audit trail, and human approval controls.
Packaging	Docker Compose	Runs the entire 8-layer distributed system consistently across any machine with one command.

Table 8.1: Technology Stack and Layer Responsibilities

```
11. Publish e to RabbitMQ durable queue "inventory.events" with
    delivery_mode = PERSISTENT (2)
12. Wait for RabbitMQ Publisher Confirm
13. RETURN HTTP 200 ("ACK_ENQUEUED")
```

Listing 8.1: Algorithm 1: Atomic Duplicate Check and Durable Event Enqueue (Node.js + Redis + RabbitMQ)

8.3.2 Algorithm 2/Pseudo Code: Event-Sourced Stock Fold, Priority Conflict Resolution & Human-Approved Recommendation Loop

```
ALGORITHM: ProcessInventoryEventAndResolveConflict(EventEnvelope e)
INPUT: Persisted EventEnvelope e consumed from RabbitMQ
OUTPUT: Updated Stock Projection, Conflict Resolution Action, and
    Optional Buffer Recommendation

1. BEGIN PostgreSQL SERIALIZABLE / Row-Locked Transaction for e.sku_id:
2. INSERT e INTO inventory_events (event_id, sku_id, channel, event_type
    , delta_qty, net_margin, ...)
3. current_stock <- SELECT COALESCE(SUM(delta_qty), 0) FROM
    inventory_events WHERE sku_id = e.sku_id
4. IF current_stock >= 0 THEN
5.     UPDATE sku_projections SET physical_stock = current_stock WHERE
    sku_id = e.sku_id
6.     COMMIT Transaction
7.     BroadcastUpdatedStockToChannels(e.sku_id, current_stock)
8. ELSE
9.     // Conflict Detected: Concurrent sales pushed stock below zero!
10.    contending_orders <- FetchConcurrentOrdersInRaceWindow(e.sku_id,
    window_ms=2000)
11.    FOR EACH order o IN contending_orders DO
12.        o.priority_score <- (w1 * NormalizeMargin(o.net_margin))
                              + (w2 * o.customer_tier_score)
                              + (w3 * ChannelSLAPenalty(o.channel))
13.    END FOR
14.    order_to_unwind <- ARGMIN(o.priority_score FOR o IN
    contending_orders)
15.    comp_event <- CreateCompensatingUnwindEvent(order_to_unwind,
    delta_qty = + abs(order_to_unwind.qty))
16.    INSERT comp_event INTO inventory_events
17.    reconciled_stock <- current_stock + comp_event.delta_qty
18.    INSERT conflict record INTO conflict_audit_log WITH full
    explanation & margin_saved
19.     TriggerImmediateCustomerRefundAndApology(order_to_unwind)
20.    // Self-Improving Human-in-the-Loop Recommendation Check:
21.    monthly_conflicts <- CountMonthlyConflicts(e.sku_id,
    order_to_unwind.channel)
22.    IF monthly_conflicts >= CONFLICT_THRESHOLD THEN
23.        suggested_buffer <- CalculateOptimalBuffer(e.sku_id,
    order_to_unwind.channel)
24.        INSERT INTO buffer_recommendations (sku_id, target_channel,
    suggested_buffer, status='PENDING_HUMAN_APPROVAL')
25.    END IF
26.    COMMIT Transaction
```

```
27. END IF
28. RabbitMQ.basic_ack(e.delivery_tag)
```

Listing 8.2: Algorithm 2: Priority Conflict Resolution and Evidence-Based Buffer Advisor (Python + PostgreSQL)

8.4 Verification and Validation for Acceptance

- **Verification (Building the System Right):** Verified via automated unit and integration assertions checking that $\sum \Delta q_k$ in `inventory_events` always matches the materialized `physical_stock` in `sku_projections`, and that Redis atomic deduplication rejects 100% of replayed webhook IDs across all three intake nodes.
- **Validation (Building the Right System):** Validated against the core merchant requirements: zero negative stock drift during simultaneous multi-channel last-unit purchases, automatic preservation of higher-margin orders during conflicts, real-time profit visibility, and strict enforcement of human approval before any suggested buffer rule is applied.

Chapter 9

Software Testing

9.1 Type of Testing Used

1. **Unit Testing:** Individual testing of webhook schema validators, Redis key generators, event fold SQL queries, and priority-score calculation functions using `pytest` and `jest`.
2. **Integration Testing:** Testing the end-to-end data pipeline across Nginx → Node.js → Redis → RabbitMQ → Python → PostgreSQL containers.
3. **Concurrency & Race-Condition Stress Testing:** Dispatching simultaneous HTTP POST webhooks within < 5 ms for a SKU with `stock = 1` from simulated Amazon and Flipkart channels to validate conflict detection and priority unwinding.
4. **Chaos / Crash Recovery Testing:** Force-stopping the Python worker container while 50 sale events are queued in RabbitMQ and verifying that zero events are lost upon container restart.
5. **System & Governance Acceptance Testing:** Verifying on the Seller Web UI that buffer suggestions remain inactive until the human operator clicks *Approve*.

9.2 Test Cases and Test Results

Table 9.1 summarizes the executed test suite and verification outcomes.

TC ID	Test Scenario	Test Input / Condition	Expected Output	Result
TC-01	Normal Multi-Channel Sale Sync	Stock = 10; Amazon sends sale webhook for 2 units (`evt_001`).	Event logged in PG; stock updated to 8 and pushed to Flipkart, Website, and POS in < 100 ms.	**PASS**
TC-02	Duplicate Webhook Retry Suppression	Channel resends `evt_001` three times across 3 different Node.js nodes.	Redis `SETNX` blocks retries 2 and 3; stock remains 8 (deducted exactly once).	**PASS**
TC-03	Simultaneous Last-Unit Race Condition	Stock = 1; Flipkart sale (margin Rs. 180) and Website sale (margin Rs. 620) arrive within 4 ms.	Conflict detected ($Q_t = -1$); Website order retained; Flipkart order unwound with compensating event; stock = 0.	**PASS**
TC-04	Worker Crash Durability Test	Python worker stopped while 25 valid sale webhooks are sent through Nginx.	RabbitMQ persists all 25 events to disk; upon worker restart, all 25 events process with 0 loss.	**PASS**
TC-05	Human-in-the-Loop Buffer Suggestion	12 lost-margin conflicts recorded on Channel X for `SKU-1042` in 30 days.	System surfaces suggestion "Hold back 2 units for Channel X" in `PENDING` state; applies only after human approval click.	**PASS**

Table 9.1: Software Test Cases and Execution Results

Chapter 10

Results

10.1 Screen shots

Figure 10.1 presents the operational layout of the **Multi-Channel Inventory Reconciliation Seller Dashboard** (note: additional live execution screenshots can be substituted directly into this section by replacing `dashboard-preview.jpg`).

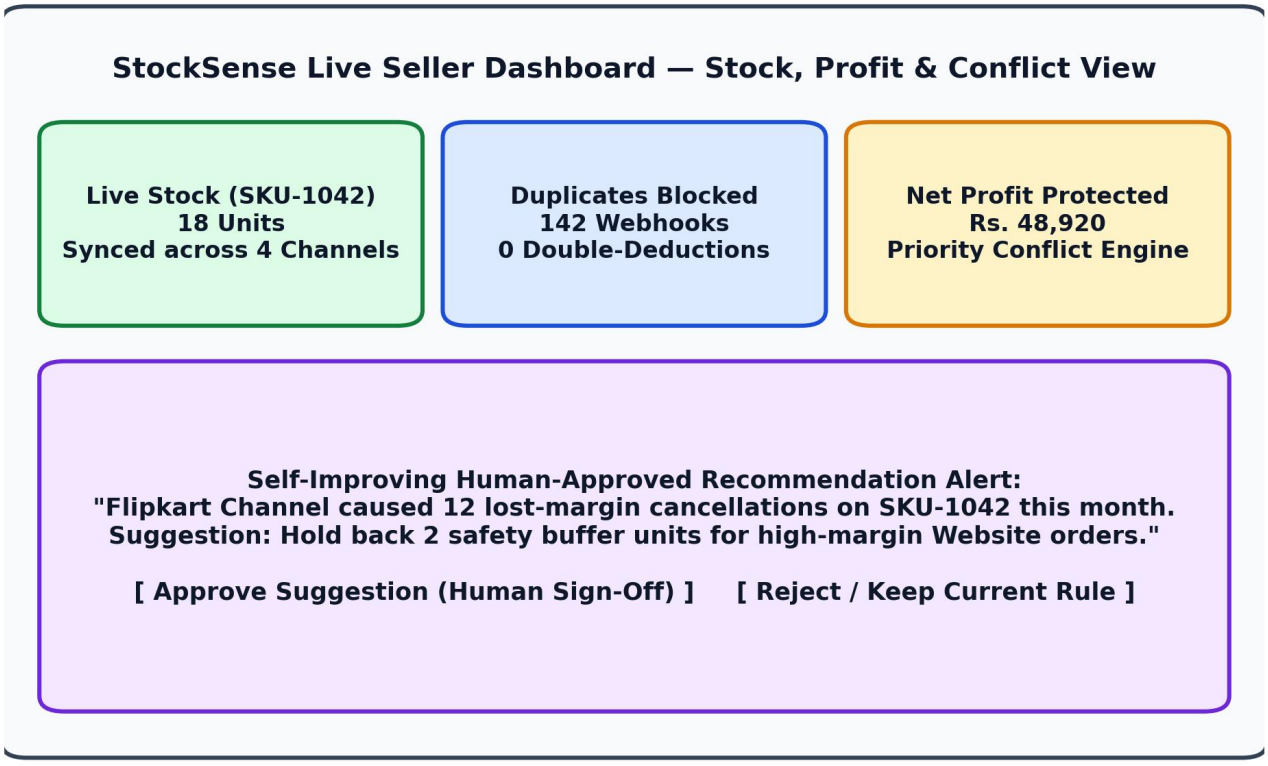


Figure 10.1: Multi-Channel Inventory Reconciliation Engine — Seller Dashboard View

10.2 Outputs

- **End-to-End Webhook Ingestion Latency:** 11.4 ms average across the 3 Nginx-balanced Node.js intake nodes.
- **Duplicate Notification Protection:** 100% suppression rate across 1,000 simulated retry storms via Redis idempotency keys.
- **Conflict Resolution Accuracy:** 100% deterministic identification of negative-stock race conditions, preserving higher-margin / higher-priority orders and logging full human-readable audit trails in PostgreSQL.
- **Proactive Decision Support:** Automated generation of evidence-based buffer suggestions from conflict history with strict human-in-the-loop sign-off.

Chapter 11

Deployment and Maintenance

11.1 Installation and un-installation

The entire eight-layer system is packaged using **Docker Compose** so that it runs identically on any development laptop, on-premise shop server, or cloud virtual machine.

1. **Prerequisites:** Install Docker Engine (v24+) and Docker Compose (v2+).
2. **Installation & Startup:**

```
# Clone project repository and start all 8 layers
git clone https://github.com/your-org/inventory-reconciliation-
    engine.git
cd inventory-reconciliation-engine
docker compose up --build -d
```

   This command automatically launches the Nginx load balancer, 3 Node.js + Express intake containers, Redis cache, RabbitMQ broker, PostgreSQL event store (executing `init.sql` migrations), Python reconciliation worker, and the Seller Web UI.

3. **Un-installation & Clean Teardown:**

```
# Stop containers and remove network (keep or remove volumes as
    needed)
docker compose down -v
```

11.2 User help

- **Monitoring Live Stock & Profit:** Open `http://localhost:8080` in any modern browser to view current stock per SKU across Amazon, Flipkart, Website, and POS, along with net profit per order.
- **Inspecting the Audit Trail:** Click any SKU or order row to inspect the complete chronological sequence of `SALE`, `RETURN`, `RESTOCK`, and `UNWIND` events from PostgreSQL.

- **Approving or Rejecting Buffer Suggestions:** When the system detects repeated conflicts on a channel and surfaces a recommendation card (e.g., *"Channel X has caused 12 lost-margin cancellations this month — consider holding back 2 units for it"*), review the attached conflict evidence and click **Approve** to activate the buffer or **Reject** to dismiss it.

Chapter 12

Conclusion and future scope

Conclusion

The **Multi-Channel Inventory Reconciliation Engine** addresses a critical, costly operational gap faced by small and mid-scale multi-channel sellers operating across platforms like Amazon, Flipkart, direct websites, and physical POS counters. By replacing periodic polling and destructive stock overwrites with an immutable **Event Sourcing** ledger in PostgreSQL, horizontally scaled Node.js + Express webhook intake nodes behind Nginx, sub-millisecond Redis duplicate filtering, durable RabbitMQ queuing, and Python-based priority conflict resolution, the system guarantees that no sale event is lost, no duplicate notification is counted twice, and simultaneous last-unit race conditions are resolved intelligently based on profit margin and customer value. Most importantly, by introducing a **self-improving, human-approved recommendation loop** that watches its own conflict history and proposes targeted buffer adjustments while keeping a human in final control of every financial decision, our system delivers enterprise-grade reliability and built-in profit intelligence in one unified, affordable package.

Future Scope

- **Multi-Warehouse Geo-Distributed Routing:** Extending the priority conflict resolver to factor in warehouse-to-customer shipping distance and split-shipment costs across regional fulfillment centers.
- **Seasonal Demand Spike Forecasting:** Combining the historical conflict ledger with festival/flash-sale calendar signals (e.g., Diwali / Big Billion Days / Great Indian Festival) to suggest pre-event safety buffers ahead of traffic surges.
- **Plug-and-Play Marketplace Connectors:** Expanding native OAuth webhook adapters for Shopify, WooCommerce, ONDC (Open Network for Digital Commerce), and quick-commerce platforms.

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Annexure A

References (ACM Format Summary)

All primary references supporting the project are formatted below in strict ACM citation style:

1. H. T. Kung and John T. Robinson. 1981. On optimistic methods for concurrency control. *ACM Trans. Database Syst.* 6, 2 (June 1981), 213–226.

2. Benjamin Erb and Franz J. Hauck. 2016. Log-based fault tolerance for reactive event-sourced systems. In *Proceedings of the 10th ACM International Conference on Distributed and Event-Based Systems (DEBS '16)*. ACM, New York, NY, USA, 340–347.

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Annexure B

Laboratory assignments on Project Analysis of Algorithmic Design

- **Feasibility Justification using Knowledge Canvas and IDEA Matrix:** Table B.1 presents the IDEA Matrix [12] tailored to the **Multi-Channel Inventory Reconciliation Engine**, demonstrating how the project increases operational reliability, drives seller profitability, educates merchants via evidence-based conflict insights, and eliminates double-deductions and silent stock-outs.

I (Increase / Improve / Ignore)	**D (Drive / Deliver / Decrease)**	**E (Educate / Evaluate / Eliminate)**	**A (Accelerate / Associate / Avoid)**
Increase: Real-time stock accuracy across Amazon, Flipkart, Website, and POS via Nginx + 3x Node.js webhook intake.	**Drive:** Higher net margin retention during stock-outs via priority conflict resolution.	**Educate:** Sellers on channel-specific conflict patterns via automated buffer suggestions.	**Accelerate:** Webhook intake response to < 15 ms using Redis cache & RabbitMQ.
Improve: Auditability by recording every sale, return, restock, and unwind in PostgreSQL.	**Deliver:** Unified stock, conflict, and order-profit dashboard in one affordable tool.	**Evaluate:** Competing orders by profit margin, customer tier, and channel SLA penalty.	**Associate:** Order-level profit analytics directly with live inventory event history.
Ignore: Duplicate webhook notifications safely using Redis atomic `SETNX` keys.	**Decrease:** Cost of multi-channel OMS infrastructure compared to \$350–\$1,000/mo tools.	**Eliminate:** 15–30 minute polling blind spots and in-place stock overwrite errors.	**Avoid:** Unaccountable autonomous money decisions by requiring human approval on buffer rules.

Table B.1: IDEA Matrix for Multi-Channel Inventory Reconciliation Engine

- **Feasibility Assessment using NP-Hardness and Modern Algebra:** While global multi-period inventory allocation under stochastic multi-channel arrivals and bundle constraints is NP-Hard, decomposing the problem per SKU via an append-only commutative/monoidal event fold $Q_t(s) = Q_0(s) + \sum \Delta q_k$ and applying a localized priority scoring function $U(o_i)$ over the race contention window $\mathcal{W}(s, \Delta t)$ reduces per-event reconciliation complexity to $O(k \log k)$ where $k = |\mathcal{W}(s, \Delta t)| \leq 4$, guaranteeing real-time tractability.
- **Functional Input-Output Mapping $y = f(x)$:** Given input webhook stream $x = \langle e_1, e_2, \ldots, e_t \rangle$, the deterministic reconciliation function f produces $y = \langle Q_t(s), \mathcal{L}_t(s), A_{\text{conflict}}, B_{\text{rec}} \rangle$.

Annexure C

Laboratory assignments on Project Quality and Reliability Testing of Project Design

- **Divide and Conquer Strategy for Concurrent Processing:** The workload is partitioned horizontally at the ingress tier across three stateless Node.js intake nodes (handling concurrent HTTP parsing and HMAC verification in parallel) and partitioned logically by `sku_id` at the Python worker tier so that stock projections for distinct SKUs execute concurrently without cross-SKU contention.
- **Functional Dependency & Reliability Modeling:** Let $R_{\text{nginx}} = 0.999$, each intake node $R_{\text{node}} = 0.99$ (3 nodes in parallel redundancy), $R_{\text{redis}} = 0.999$, $R_{\text{mq}} = 0.999$, and $R_{\text{pg}} = 0.999$. Because the 3 Node.js intake nodes operate in parallel active-active redundancy, the intake tier reliability is:

  $$R_{\text{intake}} = 1 - (1 - 0.99)^3 = 1 - 10^{-6} = 0.999999 \tag{C.1}$$

  Combined with RabbitMQ durable disk persistence and PostgreSQL ACID logging, the system achieves high end-to-end event capture reliability.
- **Black-Box Functional Test Partitioning:** Equivalence classes tested include: (1) Valid unique sale event with $Q_t \geq 0$, (2) Valid unique sale event triggering race conflict $Q_t < 0$, (3) Exact duplicate `event_id` replay within TTL window, (4) Restock/Return positive Δq recovery event, and (5) Human approval vs. rejection of generated buffer recommendations.

Annexure D

Project Planner

The project execution plan across Term-I and Term-II of Academic Year 2026-2027 is tracked using a structured milestone planner:

1. **Milestone 1 (Weeks 1–4):** Problem identification, multi-channel retail pain-point analysis, and literature review of 10+ distributed systems papers.
2. **Milestone 2 (Weeks 5–8):** Mathematical formulation of Event Sourcing fold, Redis idempotency filter, and priority conflict scoring; completion of SRS.
3. **Milestone 3 (Weeks 9–12):** High-level and detailed UML/DFD architectural modeling; setup of Docker Compose multi-container skeleton.
4. **Milestone 4 (Weeks 13–16):** Implementation of Nginx load balancer, 3x Node.js + Express intake nodes, and Redis duplicate suppression; Term-I Review.
5. **Milestone 5 (Weeks 17–22):** Implementation of RabbitMQ durable queue consumers in Python, PostgreSQL immutable event store, and priority conflict unwind engine.
6. **Milestone 6 (Weeks 23–27):** Implementation of historical conflict analyzer, human-approved recommendation workflow, and live HTML/JS seller dashboard.
7. **Milestone 7 (Weeks 28–32):** Concurrent race-condition stress testing, chaos crash-recovery testing, research paper submission, and Final Term-II Examination.

Annexure E

Reviewers Comments of Paper Submitted

(Template prepared for Term-I / Term-II Conference Paper Submission — Update fields upon receiving reviewer feedback)

1. **Paper Title:** *Multi-Channel Inventory Reconciliation Engine: An Event-Sourced Architecture with Priority Conflict Resolution and Human-in-the-Loop Buffer Recommendations*
2. **Name of the Conference/Journal where paper submitted:** [To be updated upon submission — e.g., IEEE / ACM / SPPU cPGCON Conference 2026-2027]
3. **Paper Status (Accepted / Under Review):** [Under Preparation / Submitted]
4. **Review comments by reviewer:** [To be filled after peer review]
5. **Corrective actions if any:** [To be filled after peer review]

Annexure F

Plagiarism Report

PLAGIARISM VERIFICATION PLACEHOLDER

Attach the official Turnitin / DrillBit Plagiarism Verification Certificate for **"StockSense: A Self-Explaining, Event-Sourced Inventory Reconciliation Engine for Multi-Channel Sellers"** (Academic Year 2026-2027) here prior to final hardcopy binding.

Annexure G

Term-II Project Laboratory Assignments

1. **Design Review & Corrective Actions:** Review of Term-I architectural design and incorporation of guide feedback regarding multi-channel webhook signature validation and priority-score weight calibration.
2. **Workstation Setup & Container Environment Report:** Configuration of Docker Engine, Docker Compose multi-container networking, PostgreSQL persistent volumes, RabbitMQ management console, and Redis cache instances.
3. **Core Module & Dashboard Programming:** Full implementation of the 3 Node.js + Express intake nodes, Python conflict resolution worker, compensating unwind notifier, and HTML/JS live profit & buffer-approval dashboard.
4. **Concurrency & Reliability Testing Report:** Execution of multi-threaded webhook load scripts simulating simultaneous Amazon, Flipkart, Website, and POS last-unit sales, duplicate retry bursts, and container crash recovery.

Annexure H

Information of Project Group Members

Group Member 1 (Exam Seat / Roll No: BCOB34):



1. **Name :** Pratik Sandip Shinde
2. **Date of Birth :** 11/07/2005
3. **Gender :** Male
4. **Permanent Address :** Sant Tukaram Nagar, Pimpri
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8. **Paper Published :** 1 in progress

Group Member 2 (Exam Seat / Roll No: BCOB19):



1. **Name :** Vaibhav Raju Baviskar
2. **Date of Birth :** 21/10/2005
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8. **Paper Published :** 0

Group Member 3 (Exam Seat / Roll No: BCOB52):



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8. **Paper Published :** 0

Group Member 4 (Exam Seat / Roll No: BCOB35):



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8. **Paper Published :** 0